

7.3.2 When to use Chain of Thought (CoT) prompting?

CoT prompting is useful when you want the model to demonstrate the final answer and its reasoning. It is effective for tasks that require logical deduction or problem-solving skills.

7.3.3 Example

Let us take the example of an algebra problem. In addition to asking for the solution, you will also ask for the steps taken.

ChatGPT

Prompt



Explain how you would solve the following math problem: $2x + 5 = 11$.

Response

First, I would subtract five from both sides to isolate the variable. Then, I would divide both sides by 2 to find the value of x .



In addition to learning the answer, you also wanted to learn the process. Such an application is quite helpful in educational settings. Students of various fields, not just primary or secondary, can use ChatGPT to understand how to perform multiple operations. For instance, a lawyer could ask ChatGPT to explain a particular Supreme Court judge's decision and what led them to their conclusion.

Similarly, finance students can ask ChatGPT to explain the business valuation process for private companies through an example.

This prompting method improves the transparency in AI, which has long been likened to a black box whose workings cannot be interpreted.

So, in addition to education, this technique can help understand complex decision-making processes. For instance, you could ask ChatGPT to build a machine-learning model to identify a specific viral disease based on some symptoms. You would have to submit a few examples. Then, ChatGPT could explain its process for identifying infected patients.